```java
package Ex1;

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello Hengrith!");
    }
}
```

```
Ken@Hengrith-Laptop
$  /usr/bin/env C:\\
sages -cp C:\\Users\\
3\\bin Ex1.Main
Hello Hengrith!
```

```java
package Ex2;

public class Main {
        public static void main(String[] args) {
        System.out.printf("%-15s %-15s %n", "\\n","Line break.");
        System.out.printf("%-15s %-15s %n", "\\t","Tabulation.");
        System.out.printf("%-15s %-15s %n", "\\'","Single Quote.");
        System.out.printf("%-15s %-15s %n", "\\\"","Double Quote.");
        System.out.printf("%-15s %-15s %n", "\\\\","\\ Sign.");
        System.out.printf("%-15s %-15s %n", "\\\\\\\\","\\\\ Sign.");
        System.out.printf("%-15s %-15s %n", "//","Line Comment.");
        System.out.printf("%-15s %-15s %n", "/* ... */","Block Comment.");
        System.out.printf("%-15s %n", "\"\"\"");
        System.out.printf("%-15s %-15s %n", "","Text block.");
        System.out.printf("%-15s %n", "\"\"\"");
    }
}
```

```
Ken@Hengrith-Laptop MINGW64 /d/
$  cd d:\\Github\\Semester\ 2\\
nable-preview -XX:+ShowCodeDeta
4fe4b87e5bc\\redhat.java\\jdt_w
\n                  Line break.
\t                  Tabulation.
\'                  Single Quote.
\"                  Double Quote.
\\                  \ Sign.
\\\\                \\ Sign.
//                  Line Comment.
/* ... */           Block Comment.
"""

                    Text block.
"""
```

```java
package Ex3;

public class a {
    public static void main(String[] args) {
        int height = 5;
        for (int i = 0; height >= i ; i++) {
            System.out.println(" ".repeat(i) + "*".repeat(height-i) + (((height-i-1) > 0) ? "*".repeat(height-i-1) :
""));
        }
    }
}
```

```
Ken@Hengrith-Laptop
$  /usr/bin/env c:\\
rs\\Ken\\AppData\\Ro
**********
  ********
   ******
    ***
     *
```

```java
package Ex3;

public class b {
    public static void main(String[] args) {
        int height = 3;
        int width = 3;

        for (int i = 0; height > i ; i++) {
            if (i == 0 || i == height-1) {
                System.out.println("*".repeat(width));
            } else {
                System.out.println("*" + " ".repeat(width-2) +
"*");        }
        }
    }
}
```

```
Ken@Hengrith-Laptop
$  /usr/bin/env C:\\
sages -cp C:\\Users\
3\\bin Ex3.b
***

*  *

***
```

```java
package Ex3;

public class c {
    public static void main(String[] args) {
        int height = 5;
        int width = 5;
        for (int i = 1; height >= i; i++) {
            for (int j = 0; width > j; j++) {
                System.out.print(i + j);
            }
            System.out.println();
        }
    }
}
```

```
Ken@Hengrith-Laptop
$  /usr/bin/env C:\
sages -cp C:\\Users
3\\bin Ex3.c
12345
23456
34567
45678
56789
```

```java
package Ex4;

public class DataTable {

    static int rowCounter = 0;

    static void insert(String[][] table, String... values) {
        for (int col = 0; col < values.length; col++) {
            table[rowCounter][col] = values[col];
        }
        rowCounter++;
    }

    static void printSeparator(int[] widths) {
        for (int w : widths) {
            System.out.print("+" + "=".repeat(w + 2));
        }
        System.out.println("+");
    }

    static void printThinSeparator(int[] widths) {
        for (int w : widths) {
            System.out.print("+" + "-".repeat(w + 2));
        }
        System.out.println("+");
    }

    static void printRow(String[] row, int[] widths) {
        for (int i = 0; i < row.length; i++) {
            System.out.printf("| %-" + widths[i] + "s ", row[i]);
        }
        System.out.println("|");
    }

    static int[] calcWidths(String[] headers, String[][] table) {
        int[] widths = new int[headers.length];
        for (int i = 0; i < headers.length; i++) {
            widths[i] = headers[i].length();
            for (String[] row : table) {
                if (row[i] != null && row[i].length() > widths[i])
                    widths[i] = row[i].length();
            }
        }
        return widths;
    }

    public static void main(String[] args) {

        String[] headers = { "Country", "Name", "Profession", "Age" };
        String[][] table = new String[4][headers.length];

        insert(table, "Germany", "Michael", "Computer Engineer", "19");
        insert(table, "England", "Robert", "Artist", "34");
        insert(table, "United Kingdom", "Julia", "Designer", "42");
        insert(table, "United States", "Jo", "Actor", "21");

        int[] widths = calcWidths(headers, table);

        // Print header
        printSeparator(widths);
        printRow(headers, widths);
        printSeparator(widths);

        // Print rows
        for (String[] row : table) {
            printRow(row, widths);
            printThinSeparator(widths);
        }
    }
}
```

```
Ken@Hengrith-Laptop MINGW64 /d/Github/Semester 2/Y2-S2-
$  /usr/bin/env C:\\Program\ Files\\Eclipse\ Adoptium\\
sages -cp C:\\Users\\Ken\\AppData\\Roaming\\Code\\User\
3\\bin Ex4.DataTable
+================+=========+====================+=====+
| Country        | Name    | Profession         | Age |
+================+=========+====================+=====+
| Germany        | Michael | Computer Engineer  | 19  |
+----------------+---------+--------------------+-----+
| England        | Robert  | Artist             | 34  |
+----------------+---------+--------------------+-----+
| United Kingdom | Julia   | Designer           | 42  |
+----------------+---------+--------------------+-----+
| United States  | Jo      | Actor              | 21  |
+----------------+---------+--------------------+-----+
```

```java
package Ex4;

public class MultiplicationTable {

    static int rowCounter = 0;

    static void insert(String[][] table, String... values) {
        for (int col = 0; col < values.length; col++) {
            table[rowCounter][col] = values[col];
        }
        rowCounter++;
    }

    public static void main(String[] args) {

        String[] headers = {"", "1", "2", "3", "4", "5", "6", "7", "8", "9", "10"};
        String[][] table = new String[5][headers.length];

        insert(table, "1", "1",  "2",  "3",  "4",  "5",  "6",  "7",  "8",  "9",
 "10"); insert(table, "2", "2",  "4",  "6",  "8",  "10", "12", "14", "16", "18", "20");
        insert(table, "3", "3",  "6",  "9",  "12", "15", "18", "21", "24", "27", "30");
        insert(table, "4", "4",  "8",  "12", "16", "20", "24", "28", "32", "36", "40");
        insert(table, "5", "5",  "10", "15", "20", "25", "30", "35", "40", "45", "50");

        // Print header
        for (String c : headers) System.out.printf("%4s|", c);
        System.out.println();
        System.out.println("-".repeat(headers.length * 5 + 1));

        // Print rows
        for (String[] row : table) {
            for (String cell : row) {
                System.out.printf("%4s|", cell);
            }
            System.out.println();
        }
    }
}
```

```
Ken@Hengrith-Laptop MINGW64 /d/Github/Semester 2/Y2-S2-OO
$  /usr/bin/env C:\\Program\ Files\\Eclipse\ Adoptium\\jd
sages -cp C:\\Users\\Ken\\AppData\\Roaming\\Code\\User\\w
3\\bin Ex4.MultiplicationTable
    |   1|   2|   3|   4|   5|   6|   7|   8|   9|  10|
-------------------------------------------------------
   1|   1|   2|   3|   4|   5|   6|   7|   8|   9|  10|
   2|   2|   4|   6|   8|  10|  12|  14|  16|  18|  20|
   3|   3|   6|   9|  12|  15|  18|  21|  24|  27|  30|
   4|   4|   8|  12|  16|  20|  24|  28|  32|  36|  40|
   5|   5|  10|  15|  20|  25|  30|  35|  40|  45|  50|
```

```java
package Ex5;

public class a {
    public static void main(String[] args) {
        String text = "I Love y Home Country";
        System.out.println("Text length is: " + text.length());
    }
}
```

```
Ken@Hengrith-Laptop
$  /usr/bin/env C:\\
sages -cp C:\\Users\
3\\bin Ex5.a
Text length is: 21
```

```java
package Ex5;

public class b {
    public static void main(String[] args) {
        String text = "I love my little country";
        System.out.println("Result: " +
text}substring(7));
}
```

```
Ken@Hengrith-Laptop MINGW64
$  /usr/bin/env C:\\Program\
sages -cp C:\\Users\\Ken\\Ap
3\\bin Ex5.b
Result: my little country
```

```java
package Ex5;

public class c {
    public static void main(String[] args) {
        String text = "I love my little country";
        System.out.println("\"i\" is at index: " +
text.indexOf('i'));System.out.println("\"u\" is at index: " +
text.indexOf('u'));System.out.println("\"e\" is at index: " +
text}indexOf('e'));
}
```

```
Ken@Hengrith-Laptop MINGW64
$  /usr/bin/env C:\\Program\
rs\\Ken\\AppData\\Roaming\\C
"i" is at index: 11
"u" is at index: 19
"e" is at index: 5
```

```java
package Ex5;

public class d {
    public static void main(String[] args) {
        String text1 = "Hi Students!";
        String text2 = "Students";
        String text3 = "Hi Students!";
        System.out.println("Text1 is equal to Text2: " + (text1 ==
text2));System.out.println("Text1 is equal to Text3: " + (text1 ==
text3));
}
```

```
Ken@Hengrith-Laptop MINGW64 /d/G
$  /usr/bin/env C:\\Program\ Fil
sages -cp C:\\Users\\Ken\\AppDat
3\\bin Ex5.d
Text1 is equal to Text2: false
Text1 is equal to Text3: true
```

```java
package Ex5;

public class e {
    public static void main(String[] args) {
        String text1 = "Hi Students!";
        String text2 = "Students";
        String text3 = "Teacher";

        System.out.println("Text1 contains Text2: " + (text1.contains(text2)));
        System.out.println("Text1 contains Text3: " + (text1.contains(text3)));
    }
}
```

```
Ken@Hengrith-Laptop MINGW64
$  /usr/bin/env C:\\Program\
sages -cp C:\\Users\\Ken\\Ap
3\\bin Ex5.e
Text1 contains Text2: true
Text1 contains Text3: false
```

```java
package Ex6;

class Employee {
    int employeeID;
    String name;
    char gender;
    int age;
    double salary;
    String department;

    Employee(int employeeID, String name, char gender, int age, double salary, String department) {
        this.employeeID = employeeID;
        this.name = name;
        this.gender = gender;
        this.age = age;
        this.salary = salary;
        this.department = department;
    }
}

class Car {
    String brand;
    String model;
    int year;
    String color;
    double price;
    int mileage;

    Car(String brand, String model, int year, String color, double price, int mileage) {
        this.brand = brand;
        this.model = model;
        this.year = year;
        this.color = color;
        this.price = price;
        this.mileage = mileage;
    }
}

class BankAccount {
    int accountID;
    String ownerName;
    char accountType;
    double balance;
    int pin;
    String currency;

    BankAccount(int accountID, String ownerName, char accountType, double balance, int pin, String currency) {
        this.accountID = accountID;
        this.ownerName = ownerName;
        this.accountType = accountType;
        this.balance = balance;
        this.pin = pin;
        this.currency = currency;
    }
}

class Video {
    int videoID;
    String title;
    String genre;
    double duration;
    int releaseYear;
    char rating;

    Video(int videoID, String title, String genre, double duration, int releaseYear, char rating) {
        this.videoID = videoID;
        this.title = title;
        this.genre = genre;
        this.duration = duration;
        this.releaseYear = releaseYear;
        this.rating = rating;
    }
}

public class Main {
    public static void main(String[] args) {
        Employee emp1 = new Employee(1001, "Alice", 'F', 30, 3500.50, "Engineering");
        Employee emp2 = new Employee(1002, "Bob",   'M', 45, 5200.00, "Management");

        Car car1 = new Car("Toyota", "Corolla", 2020, "White", 20000.00, 15000);
        Car car2 = new Car("Honda",  "Civic",   2022, "Black", 23000.00, 5000);

        BankAccount acc1 = new BankAccount(3001, "Charlie", 'S', 15000.75, 1234, "USD");
        BankAccount acc2 = new BankAccount(3002, "Diana",   'C', 52000.00, 5678, "EUR");

        Video vid1 = new Video(5001, "Inception",       "Sci-Fi", 148.0, 2010, 'A');
        Video vid2 = new Video(5002, "The Dark Knight", "Action", 152.5, 2008, 'A');
    }
}
```